

15_popquiz magnetic force, right hand rule

 This is a preview of the published version of the quiz

Started: Mar 22 at 7:13pm

Quiz Instructions

See my updated slides "big ideas" in dropbox



Question 1 3 pts

Consider a compass.



What is true



A compass is a horseshoe magnet.



A compass is a bar magnet. the red part is the south



a compass is a bar magnet. the red part is the North pole



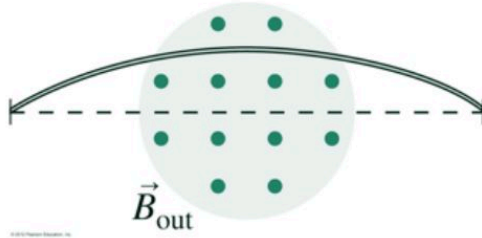
A compass points to the North of the EArth. but is not a magnet



Question 2 15 pts

- The figure shows a flexible conducting wire passing through a magnetic field that points out of the page. The wire is deflected upward, as shown. In which direction is current flowing in the wire?

- A. To the left
- B. To the right



Use the right hand rule. The dots mean that the magnetic field is toward you (out of the screen)

If you use your finger its (I, B, F)

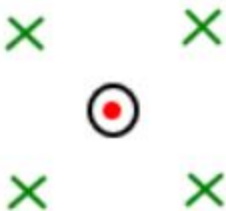
- ☐ right
- ☐ ldk
- ☐ left
- ☐ the wire will not move



Question 3 15 pts

A wire carrying current is placed in a magnetic field. The current is going toward you (out of the screen)

crosses mean that magnetic field is in the screen (so I and B are anti parallel. angle is 180). the dot means that the current is toward you (out of the screen). B is a vector



- ☐ there is no force acting on the wire



There is a force @ left acting on the wire



There is a force @ right acting on the wire

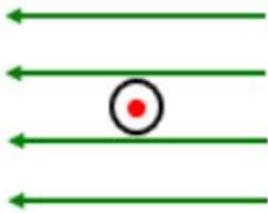


There is an up force acting on the wire



Question 4 15 pts

a wire carrying current is place in a magnetic field. The current is going toward you (out of the screen). The magnetic field is @ left. The force is (use your fight hand. I, B, F)



C



the force is @ right



upward



downward



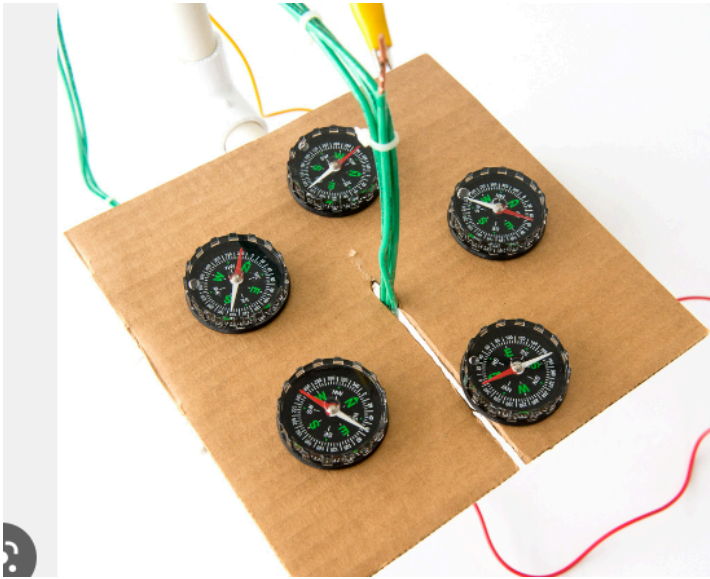
there is no force



Question 5 15 pts

A compass (red part) shows you in which direction magnetic field is flowing to. so you can think of the needle of of the compass has an arrow. the red part is the head of your arrow. and the white part the tail.

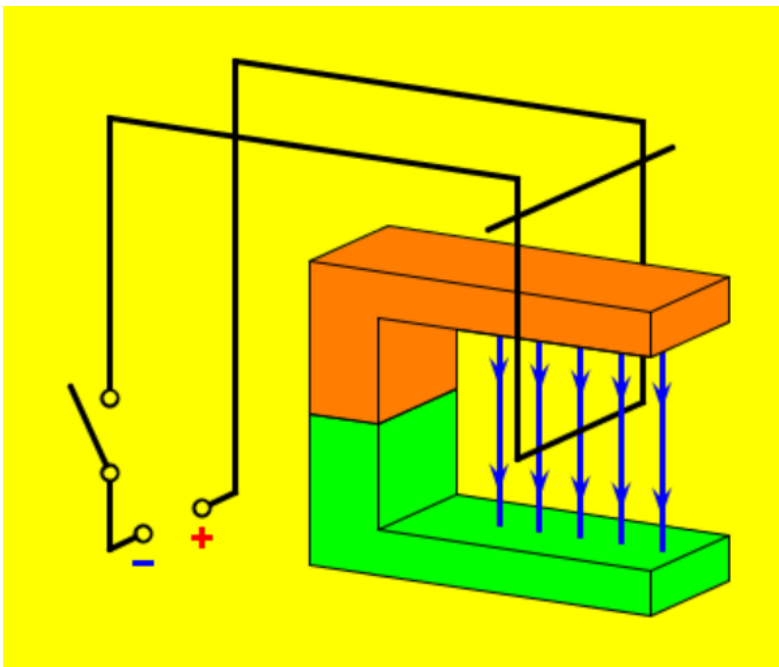
What is true? hint: use your right hand. The thumb shows the direction of the current and the curled hand which way B is "flowing" (check slide 15 of Unit3_big_ideas)



- ☐ The current is flowing down and B is curling CW
- ☐ the current is flowing up and is curling CW
- ☐ the magnetic field is diverging away from the wire and the current is down
- ☐ the current is going down and B is curling CCW



Question 6 10 pts



A piece of a wire carrying current is inside a magnetic field. (orange is the N pole).

What is true (use right hand - I, B, F). The current is toward you.



the magnetic force is up



The magnetic force is @ right



the magnetic force is @ left



the magnetic force is 0



Question 7 10 pts

A 2 meter long wire experiences a magnetic force of 8 N when placed in a 2 T magnetic field. The angle between the current and the field is 60° . Calculate the current flowing through the wire (I). round to nearest one.

hint: use equation $F = I l B \sin(\theta)$ θ is the angle between I and B.



about 5A



about 1A



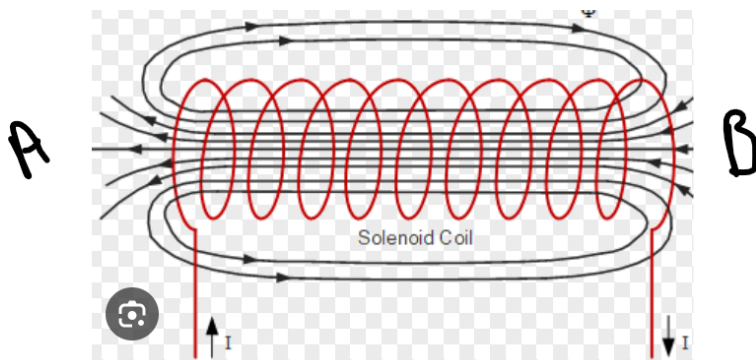
0



about 2A



Question 8 10 pts



What is true



A is a North pole



this is a monopole. A and B are a North pole

☐

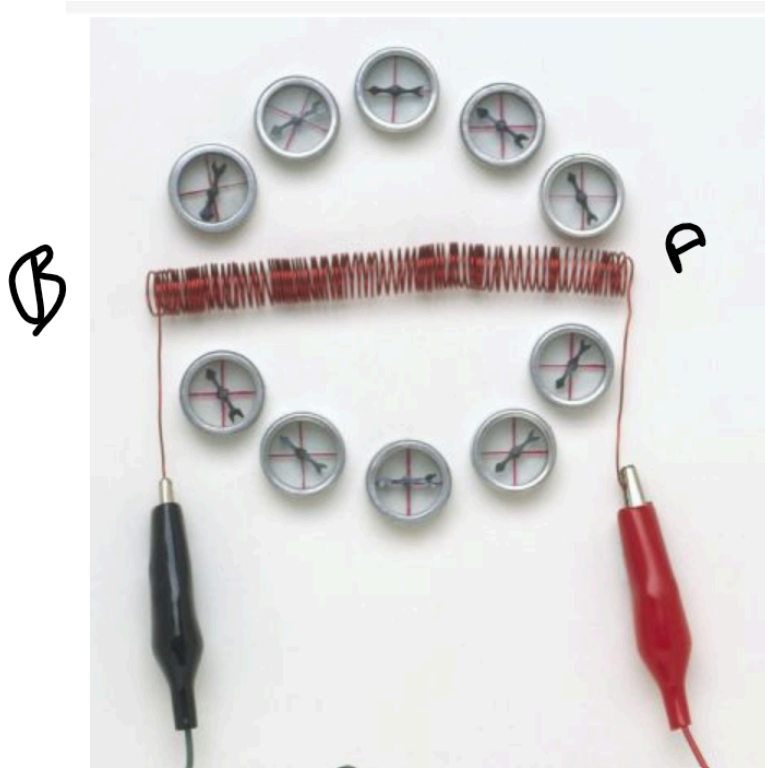
face B is a North pole.

☐

This is a coil and there is no magnetic field produced

☐

Question 9 10 pts



This coil of wire is connected to a power supply. Current is flowing. compasses are used to map out the magnetic field. Again the head is like the red part of the previous compass. the tail is like the white part of the previous compass. Use those compasses . What is true

☐

I am confused...

☐

A is the North pole

☐

B is the North pole

☐

A and B are both North pole.

Not saved

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